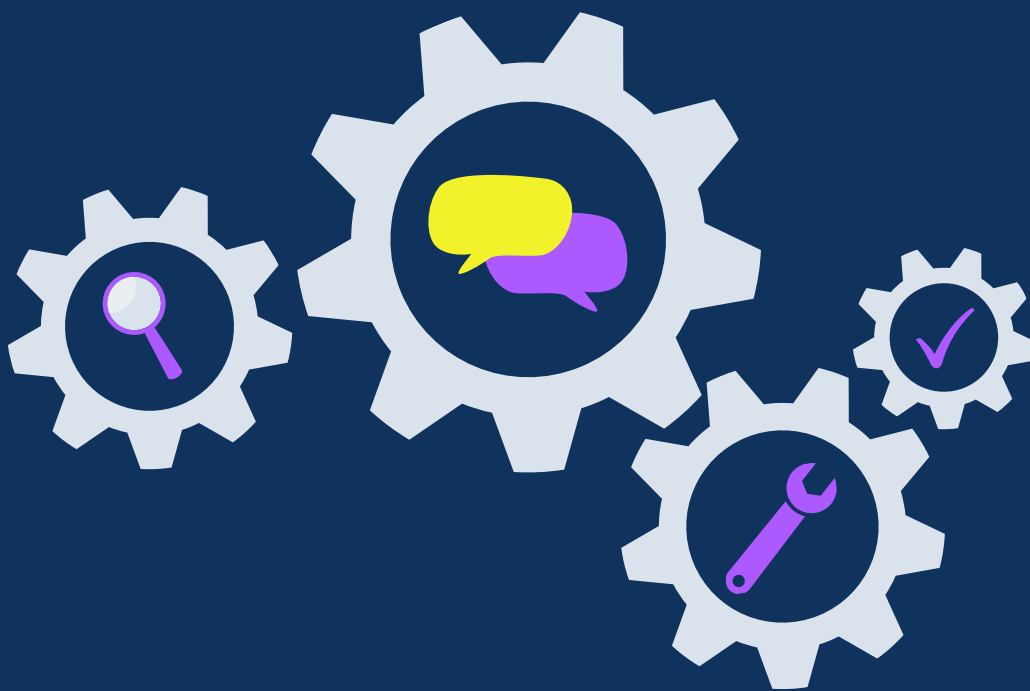


AulaSputnik

An educational project aimed at inspiring values through innovative workshops



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About AulaSputnik

A project to teach values using space as a context



Context

In the context of the new society we are building—an uncertain and complex one—it is crucial to help people develop and unlock new abilities to learn how to cooperate and participate. For this reason, a community needs to embrace and build itself through the scientific method, critical thinking, and inspiration.

The world of science and technology is a distant world for many people. This is why humanizing science and technology is a fundamental challenge, especially in the aerospace sector. The aerospace sector is a field that drives the improvement of academic and employability skills, while also innovating to find solutions to new social challenges such as climate change.

For this reason, education and the aerospace sector are two of the keys to making progress on these future collective challenges. The new generations are the key to building a more cooperative, leading, and inclusive society, where anyone—regardless of gender, background, age, or culture—can have the same opportunities.



Objetivo

AulaSputnik is an educational project focused on organizing activities designed to inspire and spark the talents of future generations, using the aerospace sector as a framework, in a way that is inclusive, accessible, and participatory for everyone.

We believe in developing scientific and technological competencies—understood as a fundamental (and therefore essential) skill—for all people, grounded in core values.

Mission

The mission is to promote positive attitudes, emotions, and identities related to science and society; the target audience is children aged 8 to 18. We will explore the importance of social participation and leadership in science and enhance academic and employability skills, using the aerospace sector as a context.

Vision

Our vision is to spark curiosity and creativity, as well as to inspire personal skills and teamwork. We aim to inspire, educate and expanding opportunities to engage with science.

Educational workshops



An innovative educational program designed to open up new skills and perspectives

Methodology

The workshop sessions follow a structure that begins with an INTRODUCTION to the CHALLENGE and leads to a CONCLUSION.

Sessions last 1.5 HOURS for groups of 15–20 people. A resource packet is provided for both participants and instructors.



List of workshops

- Challenge AulaSputnik
- Workshop 1: How do nanosatellites help our planet?
- Workshop 2: Explore the space with AulaSPUTNIK rockets!
- Workshop 3: The women in the aerospace sector
- Workshop 4: Constellations as a life notebook
- Workshop 5: Biology in space: What do aliens look like?
- Workshop 6: Create your own star system!
- Workshop 7: What do we do with the satellite data?
- Workshop 8: Build your own moon base!
- Workshop 9: What is a re-entry into a planet?



Challenge AulaSputnik

Key Features

- Age range: 8–16 years
- Type: group activity
- Difficulty: moderate
- Duration: 6–7 sessions, each lasting 1.5 hours.
- Materials included: an educational kit per workshop, LEGO building blocks, school supplies, and 3D-printed parts

Summary

Are you looking for a three-month project that integrates various STEAM activities while encouraging curiosity, creativity, and teamwork?

The AulaSputnik Challenge is the perfect activity for you!

A unique educational experience in the format of a space mission, where your students will take the lead in their own collaborative project, working as a team over several weeks to achieve a common goal.

Through interconnected workshops following a common narrative, each session builds on knowledge that is directly applied to their final space project, achieving sophisticated and valuable results for the students... and for the school as well!

Ideal for teachers looking to transform their classroom into a laboratory of ideas, collaboration, and discovery.

The students will learn...

- Constellations as a Life Notebook
- Biology in Space: What Do Aliens Look Like?
- How Do Nanosatellites Help Our Planet?
- Explore Space with AulaSputnik Rockets!
- Create your own star system!
- Egg drop (for 12-16-year-olds only)
- Women in the aerospace sector (addressed across all workshops)

Skills to be developed...

- Communication.
- Teamwork and contributing to a larger project.
- Long-term project development.
- Creativity and problem-solving skills





Educational workshops - Primary and secondary school

Workshop 1: How do nanosatellites help our planet?

The main objective of this workshop is to explain what satellites (and specifically nanosatellites) are and to explain the basic principles of this technology, so we can understand how they can help us improve our planet Earth.

Next, through a group challenge, the students must design a satellite, build a model using a basic kit, and give a presentation of their project idea. The purpose of this satellite must be to provide a service that helps our society reduce the impact of climate change.

Key Features

- Age range: 8–16 years
- Type: group activities
- Difficulty: moderate
- Duration: 1h 30 min



Workshop 2: Explore the space with AulaSPUTNIK rockets!

The main objective of this workshop is to explain what rockets are, how they work, and why they are so important for exploring space and better understanding our surroundings.

Next, through a group challenge, students will be presented with some hypotheses with the goal of designing and building a paper rocket and a launch pad. They will have to coordinate and organize themselves into roles and tasks. Afterwards, the rockets will be launched to test our initial hypotheses.

Key Features

- Age range: 8–16 years
- Type: group activities
- Difficulty: moderate
- Duration: 1h 30 min





Educational workshops - Primary and secondary school

Workshop 3: The women in the aerospace sector

The main objective of this activity is to reflect on stereotypes within STEAM professions and to give visibility to women in aerospace science and technology.

Two main activities will be carried out in this workshop. The first will focus on reflecting on and identifying unconscious behaviors and biases regarding the role of women in the aerospace sector. The second activity will involve creating a card game to learn about the career paths and professional lives of some female role models.

Key Features

- Age range: 8–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1h 30 min



Workshop 4: Constellations as a life notebook

The main objective of this activity is to understand that the night sky has played and continues to play a key role in human life, both in understanding the passage of time, organizing agricultural activity, and transmitting knowledge.

We will focus on understanding the night sky as a means of transmitting knowledge through a story, as well as on writing our own story afterward. In higher-level courses, we will learn to identify the constellations in our night sky.

Key Features

- Age range: 6–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1h 30 min





Educational workshops - Primary and secondary school

Workshop 5: Biology in space: What do aliens look like?

The main objective of this activity is to explore astrobiology and understand what life on other planets might be like based on their environment. Through a theoretical introduction and a hands-on activity, participants will discover the key concepts of the search for extraterrestrial life and how environmental conditions determine the evolution of living beings.

The workshop combines interactive explanations with a hands-on activity where students will work in groups to imagine and design an alien being based on the characteristics of a fictional planet.

Key Features

- Age range: 9–16 years
- Type: group activities
- Difficulty: moderate
- Duration: 1h 30 min



Workshop 6: Create your own star system!

The main objective of this activity is to internalize concepts such as the origin of the universe, our place within the solar system, the movements the Earth makes, the dance with our natural satellite, the Moon, and its effects, the reasons for Earth's climates and seasons, and raising awareness about the importance of caring for the environment.

The students, organized into teams and placed in a scenario exploring a new stellar system similar to our own, will have to identify and analyze the variables of the system and of the planet most likely to harbor life.

Key Features

- Age range: 11–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1h 30 min





Educational workshops - Secondary and high school

Workshop 7: What do we do with the satellite data?

What is Earth Observation? How do satellites help us monitor climate change? In this workshop, students will learn about the concept of Earth Observation and its practical applications in everyday life.

They will learn how to use tools to explore and analyze satellite images. They will work in groups to choose a topic from among various use cases related to Earth Observation and will carry out hands-on activities related to their chosen topic.

Key Features

- Age range: 12–16 years
- Type: group activities
- Difficulty: high
- Duration: 2h



Workshop 8: Build your own moon base!

In this workshop, students will discover the basic principles of space architecture and the challenges of designing a sustainable lunar base. They will reflect on key aspects such as recycling systems, pressure, and the energy required for life in space.

Using the Tinkercad tool, they will organize different modules of a base and decide which spaces are essential and which are not. They will also discuss what the interior should look like to ensure the health and well-being of the astronauts who live there.

Key Features

- Age range: 12–16 years
- Type: group activities
- Difficulty: high
- Duration: 2h





Educational workshops - Secondary and high school

Workshop 9: What is a re-entry into a planet?

How can we re-enter Earth's atmosphere—or that of another planet—from space? In this workshop, we'll explore the factors we need to consider to ensure a safe re-entry with our spacecraft.

To understand concepts such as gravity, the atmosphere, amortisation and braking systems, etc., we'll build a parachute and a landing system so that an egg can survive a free fall from a considerable height. Will our egg astronaut survive?

Key Features

- Age range: 12–16 years
- Type: group activities
- Difficulty: high
- Duration: 2h





Feedback from the workshops

Students' answers to the question:

What have you learned in the AulaSputnik workshops?

- “NANOSATELLITES CAN HELP THE WORLD.”
- “WHAT IS STEAM.”
- “MORE WOMEN SHOULD WORK IN STEAM.”
- “YOU CAN PURSUE WHAT YOU LOVE, WHETHER YOU’RE A MAN OR A WOMAN.”

Teachers' responses to the question:

What has been the most meaningful learning experience for you from the AulaSputnik workshops?

- How the STEAM concept is part of our daily lives without us even realizing it.
- The most meaningful lesson has been learning about women in the world of science and the value of women’s work in various fields.
- Reflecting on our behavior and becoming aware of our attitudes.

Space **inspires**
us and makes
us **dream!**



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